Supplementary material for: Neither conduction nor a fixed delivered fraction reproduces sixteen two-cell propagation tests: a pre-registered study

Siddharth Satte

Independent researcher, Pune, India

Every record, script and decision named here is in the study's repository at version 1.8.0 [1]; the decisions log is `DECISIONS.md`, and an entry is cited by its number (D1 to D126).

S1. The digitisation of the published histories

No raw temperature series of the two-cell tests is published: the Battery Failure Databank holds one row of summary values per single-cell test and no multi-cell test, and the article's data statement offers the data on request (D102 (1)). The supplement of Fransson et al. [2] draws the thermocouple records of two tests in its Figure 10: (a) PT_Exp4, which propagated, and (b) PP_Exp3, which did not. The figure is a vector drawing, each of its four curves a stroked path of 7,359 to 8,307 points, so the series were read out of the paths rather than off pixels.

The reading (D102 (2), `inputs/fransson2024/extract_fig10.py`) was made twice, by separate code and separate calibrations. The first uses the path objects of pdfplumber, with each panel's axes fitted by least squares to its 21 vertical and 8 horizontal major grid lines, labelled by the tick labels read as text; the second parses the figure's drawing operators with pypdf and takes the axes from the plot area's frame. Each reading is resampled every 0.5 s from 0 to 500 s. The two differ by at most 0.0005 K on PT_Exp4's neighbour curve and 0.0002 K on PP_Exp3's; the figure's resolution, half the curves' stroke width, is 2.47 and 2.46 K and 0.45 s, and the recorded uncertainty is the larger of the two. The time axis starts at the trigger's runaway, from which the paper counts the time to propagation; the figure marks PT_Exp4's onset at 271.50 s (Table 3 of the paper: 272 s).

A check (V1 of Stage 10, D102 (5)) rendered the page at 1200 dpi without anti-aliasing, calibrated the axes on the grid lines found in the pixels and traced the dashed curve column by column, at 164 and 188 whole seconds: every pixel reading lies within the recorded uncertainty plus one pixel, at most 3.03 and 2.45 K from the vector reading. PT_Exp4's record: 32.2 °C at 0 s; a spike to 102.0 °C at 3.1 s as the trigger runs away, back to 79.9 °C at 8.4 s; 141.0 °C at 45 s; a slow rise to 197.3 °C at 271.50 s; then the jump. The main text reads the first 8.5 s as the thermocouple's exposure at the trigger's runaway rather than the can's temperature (Fig. 7a). PP_Exp3's plotted maxima fall 2.9 and 14.6 K short of the paper's table, more than the figure's resolution, which the paper does not explain.

S2. The predictions of Stages 1 to 12

Each stage's predictions file was committed and pushed before any of the stage's runs, with the expectation held for each prediction; Table S1 lists all 74, shortened from the files (`manuscript/data/predictions_by_stage.csv`, which gives the file and line of each). Through Stage 10, 64 were committed and 32 failed; Stage 11 adds six, of which four failed, and Stage 12 four, all of which held.

A failure is counted as foreseen when the predictions file, before the runs, recorded it as expected, as already fixed by earlier records or an energy balance, or as following from a condition it named that then held; a hedged "may fail" does not count. By that rule 21 of the 32 failures through Stage 10 were foreseen, and one of Stage 11's (P4, fixed by Stage 7's runs at the low contact). The stages' own texts label 17 of the 21 so; the other four are Stage 3's P3 and P6 ("expected to fail" in the file's central estimates), Stage 6's P4 (fixed by an energy balance, where the texts counted Stage 5's records only) and Stage 9's P6 (failing under the condition the file named, on which the note and the README differ). Counting the three "may fail" expectations too (Stage 4's P4 and Stage 5's P1 and P6) would give 24.

Table S1: Every prediction, as its predictions file states it (shortened), with the expectation recorded before the runs and the verdict. Foreseen: a failure the predictions file recorded before the runs as expected to fail, as already fixed by earlier records or an energy balance, or as failing under a condition it named that then held. The verdicts of Stages 11 and 12 are their scorecards'.

Stage		Prediction	Expectation	Verdict
1	P1	At 33.2 kJ and 500 mW/K, for both release durations, the hottest node reaches 0.15 K/min before the lumped node does, by more than half of the lumped time.	Central estimate: 4.2 s against 29 s for the 10 s release (86 % earlier) and 11.5 s against 39 s for the 30 s release (71 % earlier).	held
1	P2	The resolved mean temperature reaches 0.15 K/min in exactly 10 of 42 cases: 33.2 kJ at 100, 200 and 500 mW/K, 21.5 kJ at 200 and 500 mW/K, both releases.	none recorded	held
1	P3	The smallest busbar conductance at which the hottest node reaches 0.15 K/min while the lumped node does not is 100 mW/K (21.5 kJ, both release durations).	Central estimate: 100 mW/K.	held
1	P4	The hottest node reaches 10 K/min in at least one case in which the lumped node does not reach 1 K/min.	none recorded	held
2	P5(a)	At 33.2 kJ and 500 mW/K, the reactive resolved cell's hottest region enters the internal-short band (255.7 C) while the reactive lumped cell does not.	P5(a) is therefore expected to fail on its second half (probability that the lumped cell stays out of the band: about one in three).	held
2	P5(b)	At 33.2 kJ and 200 mW/K, the hottest region is pushed into the internal-short band (255.7 C) by its own reaction heat, while the reactive lumped cell stays out.	The lumped half is near certain; the resolved half is about even odds.	failed

Stage		Prediction	Expectation	Verdict
2	P5(c)	At 21.5 kJ, 500 mW/K and 30 s release, neither the hottest region nor the reactive lumped cell enters the internal-short band (255.7 C).	none recorded	held
3	P1	At 33.2 kJ, 30 s release and 500 mW/K, the hottest node reaches 0.15 K/min before the lumped node in more than 95 % of kinetic draws.	Brief's central estimate: 99 %. Study's central estimate: 100 %.	held
3	P2	At 33.2 kJ, 30 s release, 500 mW/K, the hottest-node onset's central 95 % interval is narrower than 20 s and the lumped node's wider than 20 s.	Brief's central estimate: 8 s and 40 s. Study's central estimate: 9.2 s and 64 s.	held
3	P3	The smallest busbar conductance at which only the local chemistry starts is 100 mW/K in more than half the kinetic draws, and never above 200 mW/K.	Brief's central estimate: 70 % at 100 mW/K. Study's central estimate: 18.5 % at 100 mW/K, and 10.5 % at 500 mW/K: expected to fail on both counts.	failed, foreseen
3	P4	At 33.2 kJ, 30 s release, 500 mW/K, the hottest-node onset time's thermophysical standard deviation exceeds its kinetic one, ratio above 1.5.	Brief's central estimate: 3. Study's central estimate: none made.	failed
3	P5	At 33.2 kJ, 30 s release, 500 mW/K, the lumped onset time's kinetic standard deviation exceeds its thermophysical one, ratio above 1.5.	Brief's central estimate: 2. Study's central estimate: none made.	held
3	P6	Fewer than 10 % of kinetic draws show the source paper's early solid electrolyte interphase outlier behaviour at the hottest node, an onset before the node reaches 100 C.	Brief's central estimate: below 10 %. Study's central estimate: 36.5 %: expected to fail.	failed, foreseen

Stage		Prediction	Expectation	Verdict
4	P1	The anode-electrolyte submodel (holding the solid electrolyte interphase) carries over half the variance of the onset lead, lumped minus hottest-node onset, at 33.2 kJ, 30 s release, 500 mW/K.	Brief's central estimate: 65 %. Study's expectation: Held, by a wide margin.	held
4	P2	The three submodels' main-effect shares of the onset lead's variance at 33.2 kJ, 30 s release, 500 mW/K sum to between 0.8 and 1.1 (interaction small).	Brief's central estimate: none given. Study's expectation: Uncertain, no number.	held
4	P3	With the anode-electrolyte submodel (holding the solid electrolyte interphase) at its posterior mean, the share of draws whose hottest node starts below 100 C falls below 5 %.	Brief's central estimate: none given. Study's expectation: Held.	held
4	P4	Restricting the dominant submodel to the half of its posterior samples nearest the mean narrows the onset lead's 2.5 to 97.5 % width by more than 40 %.	Brief's central estimate: none given. Study's expectation: May fail.	failed
4	P5	The ordering of the three submodels by their share of the onset lead's variance is the same at 100 mW/K as at 500 mW/K.	Brief's central estimate: none given. Study's expectation: Held, if the anode-electrolyte file leads at 500 mW/K.	held
5	P1	At the central contact conductance with the posterior-mean kinetics, the resolved neighbour propagates in series and parallel, its time to propagation within the measured 147 to 342 s.	Study's expectation: May fail: by the estimates above the hot spot stays below 255.7 C, or, if the line of contact rises further, reaches it within the first minute.	failed

Stage		Prediction	Expectation	Verdict
5	P2	In both connections, series and parallel, the lumped neighbour propagates more than 20 s later than the resolved one, or not at all within 900 s.	Study's expectation: Held, most likely as "not at all": its passive peak should lie near 145 to 170 C.	held
5	P3	At the resolved model's propagation time, its mid-surface temperature lies between 150 and 255.7 C: the hot spot leads the surface a thermocouple sees.	Study's expectation: Toward the trigger, held if propagation occurs (that face is the line of contact); away from it, may fail below 150 C.	failed
5	P4	The resolved model's mean mid-surface heating rate over the 60 s before propagation lies between 0.2 and 0.8 C/s (measured about 0.4).	Study's expectation: Uncertain.	failed
5	P5	The predicted series and parallel times to propagation differ by less than 30 s (the measured means differ by 8 s).	Study's expectation: Held if both propagate: the tabs carry 4 to 8 % of the contact path.	failed
5	P6	The share of the 4,800 factorial runs in which the resolved neighbour propagates lies between 30 and 90 % (measured 9 of 16, 56 %).	Study's expectation: May fail low: the model lacks the vent, ejecta and flame heat, and the estimates above fall short of 255.7 C at the central contact.	failed

Stage		Prediction	Expectation	Verdict
5	P7	The contact and tab conductance range carries more of the spread in predicted time to propagation than the kinetic posterior (standard deviation ratio above 1), reversing Stage 3's onset finding.	Study's expectation: Held if enough runs propagate to give both spreads: the contact range changes the path 4.6-fold.	failed
6	P1	In Stage 5 runs the lumped neighbour first reaches 150 C within the measured 147 to 342 s in over half its crossings; the hot spot always before 147 s.	Brief's expectation: Likely; the Stage 5 central case already shows it. Study's expectation: Fails on its first part.	failed, foreseen
6	P2	The face away from the trigger, at the thermocouple's position, exceeds 133 C, the largest neighbour maximum of five non-propagating tests, in fewer than half the Stage 5 runs.	Brief's expectation: Uncertain. Study's expectation: Fails.	failed, foreseen
6	P3	With the trigger duration matched to the measured trigger-to-peak times, the lumped neighbour's peak rises above its Stage 5 central value (159 to 162 C) by at least 20 K.	Brief's expectation: Likely; a longer release keeps the neighbour heated past its Stage 5 peak time. Study's expectation: Fails.	failed, foreseen
6	P4	With the trigger energy raised to the Battery Failure Databank's total including the ejected fraction, the model's trigger peak falls inside the measured range of the sixteen trigger peaks.	Brief's expectation: Uncertain; the Databank's split and the tests' cells need not agree. Study's expectation: Fails.	failed, foreseen

Stage		Prediction	Expectation	Verdict
6	P5	In the case matched to the tests in duration and energy, the resolved hot spot reaches 255.7 C in at least one run, and the lumped neighbour in none.	Brief's expectation: Close to even on the first part; likely on the second. Study's expectation: Fails on its first part; holds on its second.	failed, foreseen
6	P6	In the case matched to the tests, over the 200 kinetic posterior draws, the hot spot reaches 255.7 C in between 10 and 70 % of them (central 35 %).	Brief's expectation: Uncertain. Study's expectation: Fails, at 0 %.	failed, foreseen
6	P7	In no case does the lumped neighbour reach 255.7 C before the hot spot does.	Brief's expectation: Likely. Study's expectation: Holds: where both reach 255.7 C (at the total energy, if anywhere), the line of contact leads.	held
7	P1	At the central contact with per-test triggers, C1 (hot spot at 255.7 C) predicts propagation in fewer than 9 of the 16 tests.	Brief's expectation: Likely; Stage 6 ignited the hot spot only at the Databank's total energy and, at the median peak, only at the high contact. Study's expectation: Holds, with eight.	held

Stage		Prediction	Expectation	Verdict
7	P2	At the central contact, C3 (lumped neighbour at 150 C) predicts propagation in 9 or more of the 16 tests and gets at least 10 of 16 outcomes right.	Brief's expectation: Uncertain; the paper's own criterion on the paper's own trigger peaks. Study's expectation: Fails on its second part, by one test.	failed, foreseen
7	P3	Among the propagated tests that C3 predicts, the predicted crossing time lies within 60 s of the measured propagation time in at least half.	Brief's expectation: Uncertain. Study's expectation: Fails.	failed, foreseen
7	P4	C4 (separator melting at the hot spot, 138.5 C) predicts propagation in more tests than were measured (more than 9 of 16) at the central contact.	Brief's expectation: Likely; the hot spot passed 138.5 C in every Stage 5 run at the central and high contacts. Study's expectation: Holds.	held
7	P5	An aged solid electrolyte interphase lowers the hot spot's first self-heating temperature but changes no C4 or C5 verdict at the central contact.	Brief's expectation: Likely, conditional on A4. Study's expectation: Fails on its first part; its second part is expected to hold.	failed, foreseen
7	P6	Ranking the sixteen tests by trigger energy ranks the neighbour's lumped peak in the same order within each connection (no inversions).	Brief's expectation: Likely; a check on the mapping, not a result. Study's expectation: Holds.	held

Stage		Prediction	Expectation	Verdict
7	P7	The series and parallel halves receive the same predicted propagation count within one, for every criterion, at the central contact.	Brief's expectation: Likely; the model distinguishes the connections by a tab only, and the measured counts (5 of 8 and 4 of 8) differ by one. Study's expectation: Fails, on C1 (and C6).	failed, foreseen
7	P8	In the internal-short event runs at the central contact, the resolved neighbour's peak after the event lies within the propagating neighbours' measured 381 to 763 C in at least half.	Brief's expectation: Uncertain. Study's expectation: Fails, with no run inside.	failed, foreseen
8	P1	Of the nine propagated tests, f*(C1) is 0 in the six Stage 7 hits and below 0.3 in the other three.	Brief's expectation: Likely; the hot spot is close to 255.7 C in several misses. Study's expectation: Holds.	held
8	P2	Of the seven non-propagated tests, at least three, including the two Stage 7 false alarms at 0, have f*(C1) below 0.3, so the propagated and non-propagated tests' f*(C1) ranges overlap.	Brief's expectation: Likely. Study's expectation: Holds, with all seven below 0.3.	held
8	P3	f*(C2) lies between 0.05 and 0.5 in at least twelve of the sixteen tests.	Brief's expectation: Uncertain. Study's expectation: Fails.	failed, foreseen

Stage		Prediction	Expectation	Verdict
8	P4	With 0.2 of the ejected energy delivered, C2 predicts propagation in at least nine tests, crossing within 60 s of the measured time in at least four of its hits.	Brief's expectation: Uncertain; Stage 7 had one of eight within 60 s for C3 at f = 0. Study's expectation: Fails on its second part.	failed, foreseen
8	P5	At each common fraction (0.1, 0.2 and 0.3), no criterion scores more than 12 of 16: delivered heat moves the predicted count but not past the cap the outcomes set.	Brief's expectation: Likely. Study's expectation: Holds.	held
8	P6	Within each connection, f*(C1) and f*(C2) never increase with trigger energy (no inversion).	Brief's expectation: Likely; a check on the mapping. Study's expectation: Holds, with ties.	held
8	P7	Joule heat of 0.2 of the stored electrical energy (the electrical bracket) lowers f*(C1) of the parallel tests by less than 0.1 in every test.	Brief's expectation: Uncertain. Study's expectation: Fails, on M50_PT_Exp1.	failed, foreseen
8	P8	Delivering through the quarter of the facing half nearest the cap lowers f*(C1) in every sensitivity test where f* is above 0 and raises f*(C2) in none.	Brief's expectation: Likely for the first part, uncertain for the second. Study's expectation: Holds.	held

Stage		Prediction	Expectation	Verdict
9	P1	With the 21.39 s delivery, C1 reproduces no propagated test's measured time within 30 s at any fraction at or above f*'s lower end.	Brief's expectation: Likely (the hot spot crosses at 11.9 to 18.5 s at the threshold's upper end). Study's expectation: Holds.	held
9	P2	With the 21.39 s delivery, C2 reproduces at least six of nine propagation times within 30 s, at a fraction between f*'s lower end and 0.1 above its upper end.	Brief's expectation: Uncertain (seven upper-end crossings lie inside the window; ST_Exp3's lies at 61.4 s). Study's expectation: Holds.	held
9	P3	Every C2 time-matching fraction found in P2 lies no more than 0.02 above the upper end of f*(C2).	Brief's expectation: Uncertain. Study's expectation: Uncertain; it turns on M50_PP_Exp1.	held
9	P4	For the five low-maximum non-propagated tests, the peak-matching fraction at the thermocouple node lies below both f*(C1) and f*(C2).	Brief's expectation: Likely. Study's expectation: Fails, on M50_ST_Exp1 and ST_Exp2, whatever the runs show.	failed, foreseen

Stage		Prediction	Expectation	Verdict
9	P5	The peak-matching fractions (five non-propagated tests) and the C2 time-matching fractions (nine propagated tests) separate: the largest of the former is below the smallest of the latter.	Brief's expectation: Uncertain; this is the stage's question. Study's expectation: Fails.	failed, foreseen
9	P6	With a 240 s constant-power delivery, the C1 time-matching fraction (hot spot at 255.7 C at the measured time) exceeds 0.2 in M50_PT_Exp4, M50_PP_Exp2 and M50_ST_Exp3.	Brief's expectation: Uncertain. Study's expectation: Uncertain; fails if M50_PT_Exp4 has no f_t.	failed, foreseen
9	P7	The bracket on the ejected-to-body energy ratio rescales every time- and peak-matching fraction by 0.75 to 1.40 and changes P5's verdict in neither direction.	Brief's expectation: Likely. Study's expectation: Holds.	held
9	P8	An aged solid electrolyte interphase changes the C2 time-matching fraction by less than 0.01 in M50_PP_Exp4, M50_SP_Exp1 and M50_SP_Exp2.	Brief's expectation: Likely (Stage 7: the aged SEI changed no verdict). Study's expectation: Holds.	held
10	P1	Before propagation, the model under the 21.39 s delivery, at one end of its threshold bracket, fits M50_PT_Exp4's measured history at the Stage 9 (far-face) node within 25 K root-mean-square.	Brief's expectation: uncertain.	held
10	P2	The long-delivery readings (240 and 360 s) fit M50_PT_Exp4's measured history with a smaller root-mean-square difference than the short-delivery reading.	Brief's expectation: uncertain; the measured history's shape decides, and A0.2 is recorded before this prediction is scored, not before it is written.	failed

Stage		Prediction	Expectation	Verdict
10	P3	M50_PT_Exp4's measured history rises over 30 K in the 60 s before propagation, which the short delivery reproduces within a factor of two and the 360 s delivery does not.	Brief's expectation: uncertain.	failed, foreseen
10	P4	The peak-matching fraction is 0 at all three nodes in the same four low-maximum tests as at the Stage 9 node; M50_PT_Exp1's facing-node value is below its Stage 9 value.	Brief's expectation: likely (the far face is the coolest of the three).	held
10	P5	The separation fails at every node: the largest peak-matching fraction (five non-propagated tests) exceeds the smallest C2 time-matching fraction at the Stage 9, 90 degree and facing nodes.	Brief's expectation: likely.	failed, foreseen
10	P6	Held out, no predictor beats one probability for every test by more than 0.1 in mean log score in either split direction, and no permutation p-value is below 0.05.	Brief's expectation: likely.	held
10	P7	No ranked model quantity correlates with the measured outcome or time at p below 0.05 by permutation test, and the connection splits none of them.	Brief's expectation: likely.	held
10	P8	The survey finds at least one dataset with per-test outcomes and times for cells in direct contact, and none whose cell this study's parameters cover without an invented value.	Brief's expectation: uncertain.	failed
11	P1	For each of the four tests, f*(C2) under every kinetic draw lies within a factor of two of its posterior-mean value, or within 0.025 of it where that is 0.	Brief's expectation: likely.	failed

Stage		Prediction	Expectation	Verdict
11	P2	For each of the four tests, f*(C1) under every kinetic draw lies within a factor of two of its posterior-mean value, or within 0.025 of it where that is 0.	Brief's expectation: uncertain; the hot spot's onset is local and the kinetics set it.	held
11	P3	Under each of the eight draws, the range of the C2 threshold energy across the four tests exceeds the largest within-test band of that energy across the draws.	Brief's expectation: likely.	held
11	P4	In the four tests with zero peak-matching fraction, the far face with no delivery exceeds the measured maximum under at least three of four thermal draws and at both contacts.	Brief's expectation: uncertain; the low contact and the smallest radial conductivity are the two that could remove it.	failed, foreseen
11	P5	In each of the four tests the band of f*(C2) across the four thermal draws is narrower than its band across the four kinetic draws.	Brief's expectation: uncertain.	failed
11	P6	In at least two of six tests the posterior mean matched within 10 s, a kinetic draw at that fraction removes the crossing or shifts it over 60 s.	Brief's expectation: likely.	failed
12	P1	With E_b capped at 33.2 kJ, conduction and radiation alone meet C1 in none of the eight tests at the central contact.	Brief's expectation: likely (Stage 5 found no ignition at 33.2 kJ with the two-cell study's trigger); the file records it as fixed by Stage 6's records.	held

Stage		Prediction	Expectation	Verdict
12	P2	At the high contact the same holds.	Brief's expectation: uncertain; the file records it as close to fixed (Stage 6, with other tabs).	held
12	P3	With E_e fixed at the Databank's median, the delivered energies E* across the nine propagated tests still span at least tenfold.	Brief's expectation: likely; the file records it as fixed by Stage 9's records.	held
12	P4	With E_e fixed, the propagated and non-propagated tests' fractions still overlap (the largest f_max above the smallest f_t(C2)).	Brief's expectation: likely; the file records it as fixed by Stage 9's records.	held

S3. Mesh and step convergence

Each stage that ran the resolved neighbour halved every mesh spacing, and separately every step limit, on one run near its threshold, and accepted the configuration if the hot spot's peak rise (or a crossing time) moved by less than its acceptance (Table S2). The passive integrations of Stages 1, 3 and 4 halved the tolerances, the largest step and the evaluation grid instead.

S4. Stage 12: the trigger held to the calorimetry

Stage 12 (D122 to D124; `predictions_stage12.md`, committed and pushed at 09:30:26 UTC on 3 November 2026, 34 s before its first run) held the trigger's energies to the Battery Failure Databank's measurements for this cell [3]. Part A capped the trigger's body energy at the largest of the ten LG M50 nail tests, 33,222.9 J (row 15, RUN195), in the eight tests in which the fitted trigger meets C1 with no delivery; with the cap a run depends on the test only through its connection, so four runs, series and parallel at the central and the high contact, stand for the eight, and the two central ones repeat Stage 6's records at the same energy and release to 9.2×10^{-12} K. Part B read the four Stage 11 tests' thresholds (B1) and the nine propagated tests'

Table S2: Mesh and step convergence, by stage (the repository's verification table).

Stage and run	Mesh spacing halved	Step limits halved
1: synthetic histories and archived cases	onsets move by at most 1.1×10^{-4} of themselves	
2: headline case, 1 and 4 K limits halved		band entries move by at most 7.5×10^{-5} of themselves
3: five draws on nine cases	first times move by at most	6.7×10^{-5} of themselves
5: central case, 30 s, both connections	0.27 % and 0.31 %	0.0035 % and 0.0040 %
6: matched case, both connections	0.28 % and 0.32 %	0.0033 % and 0.0037 %
7: event run of PT_Exp4, 65.52 kJ	0.15 %	0.0008 %
8: PT_Exp1 at f = 0.1 (145,195 nodes when halved)	0.52 %	0.0021 %
9: PP_Exp3, 240 s delivery at 0.0125	0.32 %	0.0006 %

f_{t}(C2) (B2) as fractions of one ejected energy for every test, the median of the ten nail tests' ejected energies, E_{fix} = 76,741.25 J (64.16 to 89.34 kJ over the ten), in place of 3.105 E_{b}. A run depends on f and E_{e} only through the delivered energy, so a fraction of E_{fix} is a delivered energy over 76,741.25 J, and the searches continued Stage 8's and Stage 9's brackets in energy, to 0.02 and 0.005 of E_{fix} or a run within 10 s of the measured time. The ten runs are in Table S3; the checks recomputed the cap and the median from the Databank's columns, every energy account and delivery (one run cut at its C1 stop at 19.0 s, having released the pulse's share to that time), and Part C by separate code.

The four predictions held: no run meets C1 at the central contact (P1) or the high contact (P2), the hot spot peaking 63.5 to 63.8 and 29.0 to 29.3 K short of 255.7 °C; the delivered energies at f_{t}(C2) span 22.6-fold over the seven tests whose searches ended matched, from 263.5 to 5,962.4 J (P3); and PT_Exp1's $f_{\max}$, 0.0272 of E_{fix}, lies above SP_Exp3's f_{t}(C2), 0.0034 (P4). The predictions file recorded P1, P3 and P4 as fixed by Stage 6's and Stage 9's records before the runs, and P2 as close to fixed; they are counted as committed predictions all the same (Table S1).

Table S3: Stage 12. (a) The capped trigger: 33.2 kJ released over 21.39 s, no delivery, the posterior mean, the central tabs, to 900 s; with the cap a run depends on the test only through its connection, so each run stands for the tests of its connection among the eight in which the fitted trigger meets C1 with no delivery. (b) The thresholds of the four Stage 11 tests as fractions of $E_{\mathrm{fix}} = 76.74$ kJ, with the delivered energy $E^* = f^* E_{\mathrm{e}}$ beside Stage 8's; Stage 8's brackets are fractions of $3.105\,E_{\mathrm{b}}$. (c) $f_{\mathrm{t}}(\mathrm{C2})$ of the nine propagated tests as fractions of E_{fix}, with the delivered energy and the run's crossing; "threshold": the search ended with no crossing at the bracket's lower end, the bracket given.

(a) Part A: the trigger capped at 33.2 kJ

Connection	Contact	Tests	Hot spot peak (°C)	Lumped peak (°C)	Far face peak (°C)	Trigger peak (°C)	C1, C2
series	central	6	192.16	159.76	133.37	531.7	none, none
parallel	central	2	191.90	161.75	135.96	531.2	none, none
series	high	6	226.69	175.92	140.15	526.9	none, none
parallel	high	2	226.39	177.40	142.42	526.4	none, none

(b) B1: thresholds

Test	Criterion	Stage 8	Stage 12, of E_{fix}	E^*, Stage 8 (kJ)	E^*, Stage 12 (kJ)
PT_Exp1	C1	(0.15, 0.1625]	(0.0545, 0.0590]	(4.181, 4.530]	(4.181, 4.530]
PT_Exp1	C2	(0.45, 0.4625]	(0.1635, 0.1680]	(12.544, 12.892]	(12.544, 12.892]
PP_Exp3	C1	(0, 0.0125]	(0, 0.0111]	(0, 1.704]	(0, 0.852]
PP_Exp3	C2	(0.0125, 0.025]	(0.0333, 0.0444]	(1.704, 3.408]	(2.556, 3.408]
ST_Exp2	C1	0	0	0	0
ST_Exp2	C2	(0.0125, 0.025]	(0.0235, 0.0352]	(1.803, 3.607]	(1.803, 2.705]
SP_Exp3	C1	0	0	0	0
SP_Exp3	C2	0	0	0	0

(c) B2: $f_{\mathrm{t}}(\mathrm{C2})$

Test	Measured (s)	Stage 9	Stage 12	$f_{\mathrm{t}}(\mathrm{C2})$, of E_{fix}	Energy (J)	Crossing (s)
PP_Exp2	147	matched	matched	0.0777	5,962.4	143.9
SP_Exp2	149	matched	matched	0.0493	3,783.4	148.5
PP_Exp1	152	matched	matched	0.0388	2,974.4	143.5
ST_Exp3	177	threshold	threshold	(0.1557, 0.1578]	(11,950.0, 12,111.5]	
SP_Exp3	210	bracketed	matched	0.0034	263.5	209.4
SP_Exp1	223	matched	matched	0.0355	2,725.6	219.0
PT_Exp4	272	threshold	threshold	(0.0607, 0.0632]	(4,656.9, 4,851.0]	
PP_Exp4	275	matched	matched	0.0237	1,819.0	268.9
ST_Exp4	342	threshold	matched	0.0154	1,184.0	337.6

S5. The energy accounts

Every run's energy account closes: the heat taken in through the boundary paths and the reaction heat (and, from Stage 7, the event and the delivered energy) equal the change in stored heat to the residuals of Table S4, against an acceptance of 10^{-9} of the heat taken in. Every delivery equals $f\,E_{\mathrm{e}}$, or the pulse's integral up to a run's stop.

Table S4: The largest energy residual of each stage's runs, as a fraction of the heat taken in.

Stage	Runs	Largest residual
2	7	1.35×10^{-12}
3 (thermal draws)	30	6.3×10^{-12}
5	5,248	8.5×10^{-12}
6	506	8.5×10^{-12}
7	186	3.42×10^{-12} (resolved), 4.34×10^{-12} (lumped)
8	230	1.97×10^{-11} (resolved), 3.24×10^{-12} (lumped)
9	670	6.8×10^{-12} (resolved), 4.0×10^{-12} (lumped)
10	30	2.5×10^{-12} (resolved), 3.3×10^{-12} (lumped)
11	216	6.1×10^{-12} (resolved), 3.6×10^{-12} (lumped)
12	10	1.2×10^{-12} (resolved), 1.0×10^{-12} (lumped)

S6. The delivery's brackets

The registered delivery is a triangular pulse of $f\,E_{\mathrm{e}}$ over 21.39 s through the half of the neighbour's can facing the trigger. Stage 8 ran its brackets on the three tests nearest the median trigger energy and on PT_Exp1 (D79, not registered). A delivery over 10 s lowers PT_Exp1's f^{*}(C1) to $(0.1, 0.1125]$, and one over 60 s raises it to $(0.275, 0.2875]$; delivering the same energy through the quarter of the facing half nearest the cap lowers it to $(0.0375, 0.05]$. The ratio's two ends scale every f^{*} by 0.75 to 1.40. In the eight parallel tests, Joule heat of 20 % of the stored 65.52 kJ over the trigger's release takes every hot spot to 255.7 °C with no delivery (D80). With the gas layer between the roll and the can of Hoelle, Zimmermann and Hinrichsen [4] (their Eq. 28 and 29, Tables V and VI), at $f = 0.2$ in the three tests nearest the median trigger, the hot spot peaks at 192.4 to 196.5 °C and does not reach 255.7 °C within 900 s, where without the layer it does at 3.5 to 3.7 s, while the can reaches 1,886 to 1,924 °C with no melting or venting

in the model (D82). Stage 9 delivered the same energy as a constant power over 120, 240 or 360 s from the start of the release (D93): 1.1 to 70.0 W reproduces seven or eight of the nine measured times through C2. Stage 10 bracketed the thermocouple's angle with three nodes at mid-length (D102 (4)): facing the trigger (0.33°, 0.06 mm from the line of contact along the surface), at the side (88.28°, 16.25 mm) and at the far face (180°, 33.14 mm).

S7. The survey of other datasets

Stage 10 surveyed the published datasets with per-test outcomes and times for cell-to-cell propagation in cells in contact or at a stated spacing, and asked whether this study's parameters cover their cell without an invented value: an LG M50, or a 21700 NMC cell with published thermal and kinetic parameters (`results/stage10_survey.md`, D106 (1)). It made 26 web searches and read each candidate on its publisher's or repository's page where that page could be fetched. Table S5 lists what it found. One other dataset has a cell this study's parameters cover: Read, Jones and Marco [5], seven LG M50 cells 3 mm apart. Its three propagated tests followed the trigger's sidewall rupture, the neighbours reaching 132 °C at their surface 2 to 5 s after the penetration, a path this model does not represent; the survey stopped there, and the decision on a further stage is taken separately.

S8. The decisions that fix an input

The entries of the decisions log that fix an input of a result reported in the main text, by number:

- The kinetics: the model and why (D1), the posterior mean (D3), the reactant masses (D4), the separator's heat sign (D5), the neighbour's density and heat capacity (D7), the histories integrated (D9), the reactive scheme and where the reactants sit (D13, D14).
- The posterior and the thermal draws: the 200 kinetic draws (D22), the 30 thermal draws and their published ranges (D27).
- The tests and the configuration with the cans in contact: the measurements (D33), the reading of the time to propagation (D34), propagation in the model (D35), the contact (D36), the tabs (D37), radiation, room losses, trigger and ambient (D38), the thermocouple (D39), the reaction heat (D40), the mesh (D42).

Table S5: The datasets the survey found (Stage 10, Part D).

Dataset	Cell	Arrangement	This study's parameters cover the cell
Fransson et al. 2024 [2], M50 tests (this study)	LG M50, NMC	two cells in contact	yes
Fransson et al. 2024, P42A tests	Molicel P42A, NCA	two cells in contact	no
Read, Jones and Marco 2024 [5]	LG M50, NMC 811	seven cells, 3 mm gaps	yes
Read et al. 2024 [6]	unnamed, M50 specification	34 cells, 1 mm gaps	not established
Allen et al. 2025 [7]	Molicel P45B	seven cells, 3 mm	no
Lamb et al. 2015 [8]	$LiCoO_2$, 18650 and pouch	cells in contact	no
Said, Lee, Stoliarov and Marshall 2019 [9]	$LiCoO_2$, 18650	arrays	no
Hoelle, Zimmermann and Hinrichsen 2023 [4]	prismatic	five-cell stack	no
Schöberl et al. 2024 [10]	prismatic NMC and LFP	braced stack	no
Torres-Castro et al. 2020 [11]	pouch	stacks	no

- The trigger: the release durations (D52), the Databank's energies (D53), the matched case (D54), one trigger per test (D65), the aged SEI and the event (D66 to D68), the separator's spread (D71).
- The delivery and the thresholds: the ratio, the ejected energy and the delivery (D77), the search (D78), the electrical path (D80), the common fractions (D81), the gas layer (D82).
- The readings: what the paper measured (D89), f_{t} (D91), $f_{\max}$ (D92), the long deliveries (D93).
- The history, the nodes and the held-out scores: the digitisation and the nodes (D102), Part B (D104), Part C (D105).
- The bands: the draws, the tests and the runs that exist (D111), the design (D112 to D115), the scoring rules (D116).

- The calorimetry's energies: the cap, the fixed ejected energy, the eight tests and what the earlier records hold (D122), the design (D123), the runs and results (D124).

References

[1] S. Satte, Onset kinetics on a resolved neighbour cell: where a whole-cell threshold and a local one disagree, Zenodo, software, all versions; version v1.8.0 used here (2026). doi:10.5281/zenodo.23279455.

[2] M. Fransson, J. Pfaff, L. Broche, M. Buckwell, C. Kirchner-Burles, H. T. Reid, S. Schopferer, A. Rack, D. P. Finegan, P. R. Shearing, Exploring thermal runaway propagation in Li-ion batteries through high-speed X-ray imaging and thermal analysis: Impact of cell chemistry and electrical connections, J. Power Sources 617 (2024) 234916. doi:10.1016/j.jpowsour.2024.234916.

[3] D. P. Finegan, J. Billman, J. Darst, P. Hughes, J. Trillo, M. Sharp, A. Benson, M. Pham, I. Kesuma, M. Buckwell, H. T. Reid, C. Kirchner-Burles, M. Fransson, D. Petrushenko, T. M. Heenan, R. Jervis, R. Owen, D. Patel, L. Broche, A. Rack, O. Magdysyuk, M. Keyser, W. Walker, P. Shearing, E. Darcy, The battery failure databank: Insights from an open-access database of thermal runaway behaviors of Li-ion cells and a resource for benchmarking risks, J. Power Sources 597 (2024) 234106. doi:10.1016/j.jpowsour.2024.234106.

[4] S. Hoelle, S. Zimmermann, O. Hinrichsen, 3D thermal simulation of thermal runaway propagation in lithium-ion battery cell stack: Review and comparison of modeling approaches, J. Electrochem. Soc. 170 (6) (2023) 060516. doi:10.1149/1945-7111/acd966.

[5] E. Read, S. Jones, J. Marco, The effect of sidewall rupture on the propensity for thermal runaway propagation in a small lithium-ion battery module, J. Power Sources Adv. 30 (2024) 100162. doi:10.1016/j.powera.2024.100162.

[6] E. Read, J. Mathew, S. Charmer, M. Dowson, D. Lorincz, I. Örökös-Tóth, M. Dobson, J. Marco, Performance of interstitial thermal barrier

materials on containing sidewall rupture and thermal runaway propagation in a lithium-ion battery module, J. Energy Storage 95 (2024) 112491. doi:10.1016/j.est.2024.112491.

[7] J. P. C. Allen, S. B. Jones, S. Charmer, E. Read, J. Marco, Propagation-resistant glass fibre reinforced polymer sleeves for the reinforcement of lithium-ion cells under thermal abuse conditions, J. Power Sources 660 (2025) 238522. doi:10.1016/j.jpowsour.2025.238522.

[8] J. Lamb, C. J. Orendorff, L. A. M. Steele, S. W. Spangler, Failure propagation in multi-cell lithium ion batteries, J. Power Sources 283 (2015) 517–523. doi:10.1016/j.jpowsour.2014.10.081.

[9] A. O. Said, C. Lee, S. I. Stoliarov, A. W. Marshall, Comprehensive analysis of dynamics and hazards associated with cascading failure in 18650 lithium ion cell arrays, Appl. Energy 248 (2019) 415–428. doi:10.1016/j.apenergy.2019.04.141.

[10] J. Schöberl, M. Ank, M. Schreiber, N. Wassiliadis, M. Lienkamp, Thermal runaway propagation in automotive lithium-ion batteries with NMC-811 and LFP cathodes: Safety requirements and impact on system integration, eTransportation 19 (2024) 100305. doi:10.1016/j.etran.2023.100305.

[11] L. Torres-Castro, A. J. Kurzawski, J. C. Hewson, J. Lamb, Passive mitigation of cascading propagation in multi-cell lithium ion batteries, J. Electrochem. Soc. 167 (9) (2020) 090515. doi:10.1149/1945-7111/ab84fa.